

Pure Substances and Mixtures — Mark schemes

PAPERS A TO D • 30 MARKS EACH • TUTOR COPY

Marks in square brackets show how the total is split. Almost every mark on this topic turns on one word — joined. Two kinds of atom joined is a compound and pure; two kinds side by side is a mixture. The other recurring loss is the melting-point test written by halves: "lower" without "over a range", or the reverse. The notes under each answer flag the specific mistake that question was built to catch.

Paper A — Medium

30 MARKS

- 1 (a) **atom** [3]
(b) **chemically joined** (bonded)
(c) **substances** (accept: elements or compounds)
- 2 Element: **oxygen, iron** [6]
Compound: **water, carbon dioxide**
Mixture: **sea water, air**
One mark per correctly placed substance.
Air and sea water in the compound column is the standard error — "made of more than one thing" is not the test; "joined" is.
- 3 A pure substance contains only one substance / one type of particle [1] [2]
Orange juice contains many substances (water, sugars, acids) mixed together, so it is a mixture [1]
- 4 (a) identical single circles, well spaced or touching, all the same [1] [4]
(b) identical groups of two or more different circles joined together, every group the same [1]
(c) two kinds of single circle scattered among each other, none joined [1]
(d) single circles of one kind scattered among identical joined groups [1]
The mark in (c) is lost if the two kinds are drawn touching in pairs — that is a compound. Check for it specifically.
- 5 (a) **Two** [1] [3]
(b) **Compound** [1]; two kinds of atom are joined together and every particle is the same (fixed ratio) [1]
"Mixture, because there are two kinds of circle" is the trap. Two kinds of atom in one particle is a compound.
- 6 Any two: the substances are not chemically joined; can be separated by physical means; no fixed ratio / any proportions; each substance keeps its own properties; no chemical reaction when it is made; melts over a range. [2]
- 7 (a) Grey and yellow specks visible / still looks like both [1]; a magnet pulls the iron out [1] [4]
(b) **Compound** [1]; evidence: magnet no longer attracts it / uniform grey / cannot be separated physically / a reaction (glowing) took place [1]
- 8 **Nitrogen and oxygen** [1] [3]
The gases are not chemically joined to each other [1]
so they can be separated physically / the proportions can vary / each keeps its own properties [1]

- 9 An alloy such as brass (copper + zinc), steel (iron + carbon), bronze (copper + tin) **[1]** [3]
 Named use, e.g. brass for taps and instruments, steel for buildings and cars **[1]**
 Accept any sensible solid-solid mixture with a use; award the third mark for saying why the mixture is better than the pure metal (harder, stronger, does not corrode) **[1]**

Paper B — Medium

30 MARKS

- 1 Joined: compound **yes**, mixture **no** [6]
 Fixed ratio: compound **yes**, mixture **no**
 Separated physically: compound **no**, mixture **yes**
 Keeps properties: mixture **yes**
 Sharp melting point: compound **yes**, mixture **no** (a range)
 One mark per box, six boxes marked (the seventh is given).
- 2 (a) **Compound [1]** [3]
 (b) The sodium and chlorine atoms are chemically joined **[1]**, and a compound has its own properties, different from the elements it is made from **[1]**
- 3 (a) The iron filings are pulled out / jump to the magnet **[1]**; in a mixture the iron keeps its own property of being magnetic **[1]** [6]
 (b) Nothing happens **[1]**; the iron atoms are now chemically joined to sulfur in a compound with its own (non-magnetic) properties **[1]**
 (c) In a compound the ratio is fixed **[1]**; any extra iron or sulfur is left over unreacted, it does not become part of the compound **[1]**
(c) is the discriminator. Students who think a compound is "whatever you mixed" have not separated compound from mixture.
- 4 (a) **R [1]**; it melts sharply (one temperature) and at the data-book value **[1]** [6]
 (b) **Q and S [1]**; they melt over a range **[1]**
 (c) P is pure (sharp melting point) **[1]** but is a different substance, not benzoic acid, because 118 °C does not match **[1]**
Sample S melts over 117–121, a range just below 122 — it is impure, and a strong candidate may say it is only slightly impure. P is the trap: sharp does not mean "the substance on the label".
- 5 Measure its melting point (heat slowly, note when it starts and finishes melting) **[1]** [3]
 Compare with the data-book value **[1]**
 A pure sample melts sharply at one temperature, equal to the data-book value **[1]**
- 6 Any two, with the reasoning making three marks in total: [3]
 the proportions of the gases vary from place to place (e.g. water vapour) — a compound has a fixed ratio
 the gases can be separated physically (fractional distillation of liquid air) — a compound cannot
 each gas keeps its own properties (oxygen still supports burning) — a compound has new properties
- 7 The hydrogen and oxygen atoms are chemically joined **[1]** [3]
 in a fixed ratio, so every water particle is identical **[1]**
 so there is only one kind of particle / one substance present, which is what pure means **[1]**

Paper C — Hard

30 MARKS

- 1 (a) **80 °C [1]**; the temperature stays constant at one value while it melts (sharp melting point) **[1]** [6]
 (b) **71 °C to 79 °C** (accept 71–80) **[1]**
 (c) Y is **impure** / a mixture **[1]**; because it melts over a range rather than at one temperature **[1]**, and starts melting below the pure melting point of 80 °C **[1]**
Both samples were heated at the same rate and both climb 20 °C per minute either side of melting — the impurity changed only the shape of the middle.
- 2 Impurity particles get in between the particles of the solid / disrupt the regular arrangement **[1]** [3]
 so the forces holding the pattern together are weakened **[1]**
 and less energy (a lower temperature) is needed to start breaking it apart **[1]**
- 3 Add water and stir **[1]**; filter **[1]** [4]
 Mixture: a residue (sand) is left on the paper **[1]**
 Pure: nothing is left on the paper, it all dissolves **[1]**
 Accept instead: measure the melting point — sharp for pure, a range for the mixture — for full marks if described properly.
The question hands over filtration apparatus, so the filtration test is expected, but the melting-point test is a correct and complete alternative.
- 4 (a) Salt mixed with ice makes it impure **[1]**; an impure substance melts at a lower temperature, so the melting point falls below the air temperature and the ice melts **[1]** [4]
 (b) Nothing has been heated — the temperature is unchanged **[1]**; what has changed is the **melting point** of the ice, which has dropped **[1]**
- 5 (a) **Element [1]**; only one kind of atom **[1]** [6]
 (b) **Compound [1]**; two kinds of atom chemically joined in a fixed ratio **[1]**
 (c) **Mixture [1]**; the two kinds of atom are not joined and the proportion can vary **[1]**
Brass is the one they get wrong — "it is a solid, so it must be a compound". Alloys are mixtures.
- 6 In carbon dioxide the carbon and oxygen atoms are chemically joined **[1]** [3]
 in a fixed ratio (one carbon to two oxygen), so every particle is the same **[1]**
 A mixture would have separate carbon and oxygen particles, not joined, and could be separated physically — carbon dioxide cannot **[1]**
- 7 Sea water is impure (salt dissolved in it) **[1]**; impurities raise the boiling point **[1]** [4]
 As it boils, water leaves as steam but the salt stays **[1]**, so the solution becomes more concentrated / more impure and the boiling point keeps rising **[1]**
The second half is the hard part: the impurity lowers melting points but raises boiling points, and the rising temperature during boiling is a range, just as melting over a range is.

Paper D — Hard

30 MARKS

- 1 (a) **Naphthalene [1]**; it melts sharply at one temperature which matches the data-book value **[1]** [6]
 (b) **Stearic acid [1]**; the range sits just below 69 °C, and an impurity lowers the melting point and spreads it into a range **[1]**
 (c) It melts over a range (118–122), so it is not pure **[1]**; a pure substance melts at one temperature, and reaching 122 °C at the end of a range is not the same as melting at 122 °C **[1]**
(b): a candidate who says "urea" because 61–67 is "below 133" has not understood that impurities lower the melting point by a few degrees, not by seventy.

- 2** Award one mark for each of these, up to six: [6]
 In water the hydrogen and oxygen atoms are chemically joined (bonded) together.
 They are joined in a fixed ratio — two hydrogen to one oxygen in every particle.
 So every water particle is identical: one kind of particle, which is what pure means.
 A compound has its own properties, different from its elements — water does not burn or support burning.
 A mixture would keep the properties of both gases and could have any proportions.
 The hydrogen and oxygen cannot be separated by physical means; it takes a chemical reaction.
For full marks the answer must address both halves: why compound (joined, fixed ratio, new properties, not physically separable) and why pure (one kind of particle). Award at most four if only one half is covered.
- 3** (a) A **mixture [1]**; the hydrogen and oxygen particles were separate, not joined to each other [1] [4]
 (b) The spark started a chemical reaction [1]; hydrogen and oxygen atoms became chemically joined to form water particles (a new substance, a compound) [1]
- 4** One mark for each genuine comparison, any three: [3]
 mixture: iron is attracted to a magnet / compound: not attracted
 mixture: grey and yellow specks visible / compound: one uniform grey solid
 mixture: any ratio of iron to sulfur / compound: fixed ratio
 mixture: can be separated physically / compound: needs a chemical reaction
 mixture: made by stirring, no energy change / compound: made by a reaction that gives out heat and light
- 5** (a) A **compound** (joined pairs of two different atoms) [1] [3]
 (b) Draw the black circles and the white circles as separate single circles scattered among each other, not touching in pairs [1]; because joining them means the atoms are chemically bonded, which is a compound, not a mixture of elements [1]
- 6** (i) The mistake: "pure" in chemistry has nothing to do with being natural or untouched [1] [5]
 [1]
 Correct version: a pure substance contains only one substance / one kind of particle, with nothing else mixed in [2]
 (ii) The mistake: it does not just move down — it also spreads out [1]
 Correct version: an impure sample melts over a range of temperatures, and that range starts below the pure melting point [1]
These are the two most common wrong sentences in the whole topic. If he can mark them in someone else's work, he is unlikely to write them in his own.
- 7** Every pure substance has its own fixed, sharp melting point, listed in data books [1] [3]
 so measuring it and matching the value identifies the substance [1]
 A mixture melts over a range that depends on how much impurity is present and what it is, so the range is not a fixed property of any one substance [1]